

Planetary Alignment Round 1: Multiscale Organization and Nonhuman Capacities in Ecological Systems

Search Parameters and Access Limits

This inquiry evaluates the ontological assumptions, detection methodologies, and empirical findings surrounding nonhuman capacities such as intelligence, sensing, learning, memory, communication, and agency across ecological scales. The research corpus relies on peer-reviewed literature, theoretical monographs, and critical commentaries concerning plant–fungal mutualisms, common mycorrhizal networks (CMNs), plant neurobiology, holobiont theory, and planetary homeostasis. Key search terms included "arbuscular mycorrhizal reciprocal rewards," "Mimosa pudica habituation," "fungal electrical spiking," "common mycorrhizal networks," "holobiont individuality," and "Darwinizing Gaia" alongside "Maximum Entropy Production." Sources inspected include empirical studies utilizing isotopic tracing, transcriptomics, gas chromatography-mass spectrometry, and microelectrode array recordings, as well as theoretical frameworks analyzing evolutionary selection without strict genetic replication. Access was limited to published journal abstracts, full-text excerpts, secondary scientific journalism, and theoretical synopses. Significant omissions include the raw longitudinal datasets for fungal electrical recordings and complete spatial mapping data for *in situ* CMN field trials. Consequently, the analysis relies heavily on peer-reviewed interpretations and subsequent adversarial rebuttals to weigh the rigor of the primary claims.

Multiscale Organization and Ecological Mechanisms

The following table synthesizes the observation of nonhuman capacities—including exchange, signaling, learning, memory, feedback, selection, and resilience—across ascending biological scales. It explicitly contrasts distributed and organismal accounts, identifying the specific observations that either discriminate between these frameworks or remain compatible with both.

Scale	Substrate & Boundary	Mechanisms (Exchange, Signaling, Learning, Memory)	Feedback, Selection, Resilience, & Regulation	Distributed vs. Organismal Accounts & Discriminating Observations
Plant–Fungal Partnership	Arbuscular mycorrhizae (AM); plant root interface.	Bidirectional exchange of host carbohydrates for fungal mineral nutrients (e.g., phosphorus).	Biological market selection: selective rewarding of cooperative partners and sanctioning of defectors.	Organismal: Host plant centrally enslaves the symbiont. Distributed: Bidirectional agency where both optimize exchange. <i>Discriminator:</i> Fungal restriction

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				of nutrients to "cheating" roots exclusively supports the distributed account.
Rhizosphere	Local root zone; limited mycelial connections.	Transmission of defense cues (e.g., methyl salicylate), electrical impulses, and volatile organic compounds.	Anticipatory defense priming in receiver plants enhances local survival.	Organismal: Passive diffusion of chemical leakage. Distributed: Active signaling network. <i>Compatible:</i> Both accounts accommodate the observation of defense activation; isotopic tracking along specific hyphae is required to discriminate.
Community	Common Mycorrhizal Networks (CMNs) spanning forest floors.	Inter-plant resource routing; carbon transfer between distinct plant species.	Source-sink gradients regulate nutrient flow based on light, water, and host physiological status, enhancing community resilience.	Organismal: Plants are independent competitors; CMN impacts are experimental artifacts. Distributed: Forest operates as a highly connected, cooperative architecture. <i>Discriminator:</i> Successful transfer through 35 μ m mesh barriers excluding roots favors the distributed account.
Ecosystem	Holobiont; multispecies	Intra-organismal and horizontal	Host immune system acts as a	Organismal: Insular, genetically

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	consortia of host and microbiome.	transmission of symbionts; metabolic integration across species barriers.	regulatory mechanism for managing, rather than merely repelling, microbial populations.	defined individual. Distributed: Sympoietic collective blurring host-microbe boundaries. <i>Discriminator:</i> Symbiont-directed host development (e.g., organogenesis) strongly favors the distributed account.
Earth Systems	Biosphere and atmosphere; Gaian biogeochemical cycles.	Planetary biogeochemical cycling modifying the abiotic environment (e.g., atmospheric oxygen/methane homeostasis).	Differential persistence of stable metabolic cycles selected via Maximum Entropy Production (MEP) limits.	Organismal: Teleological planetary superorganism. Distributed: "It's the song, not the singer" (ITSNTS); persistence of interaction patterns. <i>Compatible:</i> Broad atmospheric homeostasis fits both; teleological intent is inherently non-identifiable.

Building upon the scaled framework above, three open questions regarding the relationships across these scales emerge. First, how do localized, bidirectional market dynamics at the plant-fungal root interface cascade to dictate the macro-level topology and resilience of forest-wide CMNs? Second, to what extent does the immunological regulation of the holobiont mirror the cybernetic feedback loops proposed at the planetary scale, suggesting a fractal nature to multispecies integration? Third, if persistence-based selection stabilizes global biogeochemical cycles, does this same thermodynamic principle govern the long-term stabilization of specific mycorrhizal partnerships despite high species turnover in the surrounding soil?

Priority Claims and Open Questions

The first priority claim investigates bidirectional agency within arbuscular mycorrhizal symbioses, explicitly challenging the assumption that plants autonomously control their microbial partners. Proposed by Kiers et al. (2011) in *Science*, the central proposition asserts

that plant-fungal mutualisms operate as a biological market where both entities exhibit agency by detecting, discriminating, and selectively rewarding optimal exchange partners. The evidence type involves precisely tracking ^{13}C and ^{33}P isotopes alongside fungal RNA enrichment in tightly controlled, compartmentalized in vitro systems over a timescale of days to weeks. An alternative explanation is that apparent selective allocation merely represents passive, stoichiometric source-sink diffusion where chemical gradients, rather than active evaluation, dictate flow. This claim is heavily supported by experiments that manipulated carbon availability to force "cheater" behavior, resulting in active fungal restriction of phosphorus specifically to the underperforming roots. The claim would be weakened if future transcriptomic analysis revealed that fungal nutrient transporter proteins operate at constant rates regardless of the specific host's carbon contribution.

The second priority claim focuses on anticipatory defense signaling via CMNs, testing the capacity for decentralized communication. Located in Babikova et al. (2013) published in *Ecology Letters*, the proposition asserts that CMNs act as subterranean information conduits, allowing plants to detect herbivore attacks on neighboring plants and invoke chemical defenses prior to direct infestation. The evidence relies on gas chromatography-mass spectrometry detecting volatile emissions (such as methyl salicylate) and behavioral assays of parasitoid wasps. The physical boundary was established using mesocosms with $35\text{ }\mu\text{m}$ mesh barriers that restricted root contact but allowed hyphal connection, with reactions measured within 24 to 72 hours of aphid infestation. A primary alternative is that defense triggers result from unavoidable aerial volatile leakage or exudate diffusion through soil water, rather than directed hyphal transport. The claim is supported by the meticulous bagging of above-ground foliage to prevent aerial contamination, demonstrating that the signal traversed the soil matrix. However, the claim remains partially unresolved for broader ecological contexts; while the mechanism implements a signaling capacity in controlled settings, unconstrained field environments contain multiple overlapping vectors that complicate the isolation of CMN-specific communication.

The third priority claim transitions to the planetary scale, addressing the evolutionary persistence of biogeochemical cycles. Championed by Doolittle (2024) in *Darwinizing Gaia: Natural Selection and Multispecies Community Evolution*, the proposition states that Earth system homeostasis evolves through the differential persistence of biological processes ("the song") rather than just the differential reproduction of the taxa executing them ("the singers"). The evidence consists of conceptual arguments extending multilevel selection theory, bolstered by empirical metagenomic observations showing that microbial community functions remain stable despite rapid taxonomic turnover. Operating at the boundary of the planetary biosphere over geological epochs, this framework confronts the alternative that biospheric homeostasis is either an incidental byproduct of standard neo-Darwinian selection at the organismal level or a purely physical phenomenon driven by Maximum Entropy Production (MEP) without any biological selection mechanism. This claim would be significantly supported by advanced mathematical modeling proving that non-reproducing but persisting interaction cycles consistently outcompete less stable cycles over deep time. It is inherently weakened by traditional evolutionary arguments asserting that complex adaptations cannot accumulate without the high-fidelity replication provided by genetic inheritance.

Disagreement Map: Empirical and Conceptual Frictions

The evaluation of nonhuman capacities is heavily fractured by detection bias, differing thresholds for what constitutes evidence, and the continuous tension between under-recognition and over-attribution.

A profound empirical disagreement exists regarding the ubiquity and ecological relevance of

CMNs in natural forests. Karst et al. (2023) map a systemic positive citation bias, arguing that the scientific literature has increasingly over-attributed forest-wide resource sharing and inter-tree facilitation to CMNs. They highlight that field experiments frequently fail to rule out alternative transfer pathways, noting that mesh barriers often alter root-soil hydrological dynamics and that true network continuity is rarely confirmed by genetic mapping. Conversely, Simard et al. (2024) defend the CMN paradigm, arguing that laboratory reductions fail to capture emergent field dynamics and that the integration of DNA sequencing with isotopic tracing sufficiently validates network connectivity and facilitation. Here, detection bias is paramount: highly controlled laboratory environments reliably detect specific hyphal exchanges, but deploying these methodologies in dense, overlapping field networks yields highly confounded variables. The failure to detect a signal in the field is frequently interpreted as an absence of the capacity, whereas it may merely indicate an unsuitable test under chaotic conditions. Conceptual friction is highly visible in the attribution of cognitive capacities, such as learning and memory, to plants. The "plant neurobiology" framework, advanced by Baluška and Mancuso, posits that plants process information via long-distance electrical signals and vesicle-mediated auxin transport, drawing direct functional parallels to animal nervous systems. Critics vehemently argue this is a semantic over-attribution that anthropomorphizes decentralized biochemical coordination. This epistemological clash is perfectly isolated in the debate over habituation in *Mimosa pudica*. Gagliano et al. (2014) demonstrated that *Mimosa* ceased its defensive leaf-folding behavior after repeated non-harmful drops, retaining this behavioral shift for up to 28 days, which they interpreted as learned habituation. Biegler (2018) sharply contested this, arguing that the experimental design lacked rigorous dishabituation controls and that the cessation of movement could be fully explained by motor fatigue. If a plant's physical motor mechanisms are exhausted, traditional behavioral assays will falsely register an absence of cognition. Similarly, the investigation into fungal communication reveals the risks of linguistic over-attribution. Adamatzky (2022) measured the electrical spiking activity of fungi such as *Omphalotus nidiformis* and *Schizophyllum commune*, discovering that these spikes cluster into specific patterns. Information complexity analysis revealed that the distributions of these fungal "word" lengths mathematically match those of human languages, suggesting a vocabulary of up to 50 distinct signals. While the empirical recordings of the extracellular electrical potentials are robust, interpreting these clusters as a semantic language invites severe over-attribution. A mechanistic alternative posits that these spikes reflect localized ion fluxes associated with growth fronts, hydration shifts, or nutrient foraging rather than conversational intent. Recognizing this activity as a form of distributed biological telemetry preserves the organism's complex sensing capacities without forcing it into a hierarchy of human likeness.

Ontology and Boundary Note

This analysis assumes a process and relational ontology. Under this paradigm, biological boundaries are highly porous, continuously overlapping, and defined by dynamic functional interactions rather than static physical borders or strict genetic lineage. Individuality is treated as an emergent property of sympoiesis. This reveals the "holobiont"—where an animal or plant host and its diverse microbiome form a functionally integrated ecological and evolutionary unit, rendering the traditional boundaries of physiology and immunology fluid. Similarly, this relational ontology accommodates Doolittle's ITSNTS theory, revealing how persistent ecological interactions (such as the nitrogen cycle) can undergo natural selection independent of the specific physical taxa executing them.

The primary alternative is the mechanistic, neo-Darwinian organismal account, which defines individuality through strict genetic inheritance, vertical reproduction, and discrete physical

bodies. This ontology excels at identifying precise molecular mechanisms, isolating specific gene transcriptions during mycorrhizal colonization, and accurately modeling inclusive fitness and kin selection. However, it obscures the reality of horizontal gene transfer, the absolute necessity of symbionts for host ontogeny, and the emergent cooperative behaviors of multi-species networks, often forcing highly integrated mutualisms to be viewed merely as reciprocal parasitism.

Many disagreements stemming from these metaphysical commitments have testable implications. For example, whether a holobiont operates as a distinct biological individual can be tested by examining the host immune system. If the immune system actively manages and curates symbiotic microbial populations for developmental signaling rather than universally repelling non-self entities, the relational boundary is empirically supported. Conversely, whether the planetary biosphere operates as a teleological organism or a complex matrix of persistent thermodynamic cycles is largely untestable, relying heavily on conceptual arguments regarding what technically qualifies as a "unit of selection".

Next Bounded Desk-Research Step

Question: How do the quantitative thermodynamic limits established by the Maximum Entropy Production (MEP) principle mathematically constrain or enable the differential persistence of biological cycles proposed in the ITSNTS framework over deep evolutionary time?

Requirements: Answering this requires a cross-disciplinary synthesis of thermodynamic physics and theoretical biology. Specifically, it necessitates analyzing models of specific Earth system processes—such as radiative entropy fluxes and global carbon cycling—against longitudinal metagenomic data demonstrating microbial functional redundancy. The objective is to determine if biological cycles that exhibit high persistence under the ITSNTS model mathematically align with the steady states that maximize entropy dissipation. This step must strictly retain existing dissent regarding whether MEP is a heuristic statistical principle or a driving physical law of biological evolution, avoiding the manufacture of a false consensus.

*Report generated by AI Model / Version: Advanced Large Language Model (Persona-Assigned).
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Works cited

1. (PDF) Reciprocal Rewards Stabilize Cooperation in the Mycorrhizal, https://www.researchgate.net/publication/51566071_Reciprocal_Rewards_Stabilize_Cooperation_in_the_Mycorrhizal_Symbiosis 2. Reciprocal rewards stabilize cooperation in the mycorrhizal symbiosis, <https://research-portal.uu.nl/en/publications/reciprocal-rewards-stabilize-cooperation-in-the-mycorrhizal-symbi/> 3. Plants and fungi recognize generous trading partners - Science News, <https://www.sciencenews.org/article/plants-and-fungi-recognize-generous-trading-partners> 4. Hart M, Forsythe J, Oshowski B, Bücking H, Jansa J, Kiers ET, https://www.researchgate.net/publication/257799086_Hart_M_Forsythe_J_Oshowski_B_Buckin_g_H_Jansa_J_Kiers_ET_Hiding_in_a_crowd-does_diversity_facilitate_persistence_of_a_low-qu ality fungal partner in the mycorrhizal symbiosis *Symbiosis* 59 47-56 5. Underground signals carried through common mycelial networks, <https://pubmed.ncbi.nlm.nih.gov/23656527/> 6. Plant-Plant Communication Through Common Mycorrhizal Networks, https://www.researchgate.net/publication/309816702_Plant-Plant_Communication_Through_Co mmon_Mycorrhizal_Networks 7. Positive citation bias and overinterpreted results lead to ... - Gwern.net, <https://gwern.net/doc/statistics/bias/2023-karst.pdf> 8. Experimentally Testing Effects

of Mycorrhizal Networks on Plant,
https://www.researchgate.net/publication/300230258_Experimentally_Testing_Effects_of_Mycorrhizal_Networks_on_Plant-Plant_Interactions_and_Distinguishing_Among_Mechanisms 9.
 “Watch out mate, aphids are about!” Plant-to-plant communication,
<https://plantscientist.wordpress.com/2013/06/24/watch-out-mate-aphids-are-about-plant-to-plant-communication-via-mycorrhizal-fungi/> 10. Processes and patterns of interaction as units of selection - PNAS, <https://www.pnas.org/doi/10.1073/pnas.1722232115> 11. Processes and patterns of interaction as units of selection,
https://www.researchgate.net/publication/324024258_Processes_and_patterns_of_interaction_as_units_of_selection_An_introduction_to_ITSNTS_thinking 12. Darwinizing Gaia | Penguin Random House Higher Education,
<https://penguinrandomhousehighereducation.com/book/?isbn=9780262380621> 13. Molter on Doolittle, 'Darwinizing Gaia: Natural Selection and ... - H-Net,
<https://networks.h-net.org/group/reviews/20125832/molter-doolittle-darwinizing-gaia-natural-selection-and-multispecies> 14. The Maximum Entropy Production Principle - MPG.PuRe,
https://pure.mpg.de/rest/items/item_1692992_2/component/file_1692991/content 15. It is not the entropy you produce, rather, how you produce it,
<https://royalsocietypublishing.org/rstb/article/365/1545/1317/21229/It-is-not-the-entropy-you-produce-rather-how-you> 16. Adaptation by Cumulative Selection - arXiv,
<https://arxiv.org/pdf/2602.20883> 17. New Version of Natural Selection Goes Mystical – CEH,
<https://crev.info/2018/04/new-natural-selection-mystical/> 18. Opinion: Response to questions about common mycorrhizal networks,
<https://www.frontiersin.org/journals/forests-and-global-change/articles/10.3389/ffgc.2024.1512518/pdf> 19. Plant neurobiology: an integrated view of plant signaling,
https://www.researchgate.net/publication/6942772_Plant_neurobiology_an_integrated_view_of_plant_signaling 20. Plant neurobiology: From stimulus perception to adaptive behavior,
https://www.researchgate.net/publication/26882093_Plant_neurobiology_From_stimulus_perception_to_adaptive_behavior_of_plants_via_integrated_chemical_and_electrical_signaling 21. The “plant neurobiology” revolution - PMC - NIH,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11085955/> 22. Insufficient evidence for habituation in *Mimosa pudica*. Response to,
https://www.researchgate.net/publication/321652482_Insufficient_evidence_for_habituation_in_Mimosa_pudica_Response_to_Gagliano_et_al_2014 23. (PDF) Language of fungi derived from their electrical spiking activity,
https://www.researchgate.net/publication/359770532_Language_of_fungi_derived_from_their_electrical_spiking_activity 24. Language of fungi derived from their electrical spiking activity - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8984380/> 25. Do Mushrooms Communicate? A Concise, Scientific Look at Fungal,
<https://medium.com/@francesco.cozzolino/do-mushrooms-communicate-a-concise-scientific-look-at-fungal-signalling-e9cae5f918ef> 26. Symbiosis As The Way Of Eukaryotic Life: The Dependent Co,
<https://works.swarthmore.edu/cgi/viewcontent.cgi?article=1193&context=fac-biology> 27. A symbiotic view of life: we have never been individuals - PubMed,
<https://pubmed.ncbi.nlm.nih.gov/23397797/> 28. A Sympoietic View of Life: Gaia as a Holobiont Community, <https://www.preprints.org/manuscript/202309.1072> 29. UNIVERSIDAD DE GRANADA FACULTAD DE ... - DIGIBUG Principal,
<https://digibug.ugr.es/bitstream/handle/10481/35213/24464971.pdf?sequence=1> 30. Microbes Drive Evolution of Animals and Plants: the Hologenome,
<https://journals.asm.org/doi/abs/10.1128/mbio.01395-15> 31. Inter-kingdom communication and the sympoietic way of life - Frontiers,
<https://www.frontiersin.org/journals/cell-and-developmental-biology/articles/10.3389/fcell.2024.1>

427798/full 32. Review of Ford Dolittle Darwinizing Gaia H-net - ResearchGate,
https://www.researchgate.net/publication/397867048_Review_of_Ford_Dolittle_Darwinizing_Gaia_H-net 33. how entropy, work and limits shape Earth system functioning. Here,
<https://earthsystem.org/2023/09/19/new-paper-working-at-the-limit-how-entropy-work-and-limits-shape-earth-system-functioning-here-is-a-simple-but-long-summary-of-the-key-points/> 34. [PDF] The Maximum Entropy Production Principle: Its Theoretical,
<https://www.semanticscholar.org/paper/The-Maximum-Entropy-Production-Principle%3A-Its-and-Dyke-Kleidon/dd38410e9cb6e7fb37aa1581627a8a31aecac436>